

AJADICO ENERGY · ENGINEERING DOCUMENTATION

BRD VERSION 3 COMPLIANT

Filling Station Management System

Comprehensive Architectural Blueprint, Forecourt Reconciliation Protocols, 17 Screen Specifications, and Supabase / Flutter Implementation Manual

AUTHOR & ENGINEERING

Antigravity Engineering

KEY STAKEHOLDERS

Senior, Dickson, Joheobe

TARGET NETWORK

5 Retail Fuel Stations (PMS & AGO)

Table of Contents

1. Executive Summary & Operational Goals	Page 2
2. Multi-Branch & Multi-Tenant Architecture	Page 2
3. Forecourt Roles, Permissions & Hardware Form Factors	Page 3
4. The Operational Daily Cycle & Mathematical Formulas	Page 4
5. Anti-Fraud & Revenue Leakage Prevention Engine	Page 6
6. Offline-First Forecourt Resilience Strategy	Page 7
7. Complete 17-Screen Specification Matrix	Page 8
8. Supabase & PostgreSQL Production Database Architecture	Page 11
9. BRD Section 7: Key Open Business Decisions	Page 12

1. Executive Summary & Operational Goals

The Ajadico Filling Station Management System is a unified, cloud-first management platform engineered to automate forecourt fuel operations, eliminate cash and volume shrinkage, and establish strict branch profitability across five retail stations (Lekki Road, Ikeja, Victoria Island, Surulere, and Ajah Terminal).

Initially supporting **Premium Motor Spirit (PMS / Petrol)** and **Automotive Gas Oil (AGO / Diesel)**, the software operates in parallel with existing physical record books during rollout, ensuring operational continuity while establishing indisputable digital accountability.

Primary Business Mandate: Replace delayed, error-prone end-of-month accounting with instant, shift-by-shift reconciliation where every litre dispensed through pump nozzles is verified against physical bank receipts, POS terminals, and physical note counts.

2. Multi-Branch & Multi-Tenant Architecture

The platform uses a centralized relational model where each station operates as an independent accounting branch while executive management retains real-time consolidated visibility.

Branch / Station	Tank Configuration	Pumps & Nozzles	Initial Pricing
Lekki Road Station	3 Tanks (T1 PMS 45kL, T2 PMS 45kL, T3 AGO 33kL)	2 Islands, 4 Nozzles	PMS: ₦1,050 AGO: ₦1,320
Ikeja Branch	Interlocked Tanks (T1 & T2 feeding common manifold)	3 Islands, 6 Nozzles	PMS: ₦1,050 AGO: ₦1,320

Victoria Island Station	Fixed 1:1 Tank Mappings	4 Islands, 8 Nozzles	PMS: ₦1,050 AGO: ₦1,320
Surulere Station	Interlocked Tanks (T1 & T2)	2 Islands, 4 Nozzles	PMS: ₦1,050 AGO: ₦1,320
Ajah Terminal	Fixed 1:1 Tank Mappings	3 Islands, 6 Nozzles	PMS: ₦1,050 AGO: ₦1,320

3. Forecourt Roles, Permissions & Hardware Form Factors

Hardware access is tailored strictly to physical working conditions across three distinct form factors:

Role	Device & Environment	Authentication	Core Permissions & Operational Limits
Pump Attendant	Shared Station Tablet (Mounted on forecourt)	Fast 4–6 digit PIN (Zero typing)	Confirm opening meter readings; log pump credit sales; log calibration pour-backs; enter closing meter readings; submit 4-way remittance declaration; snap POS slips. Cannot alter baseline readings or delete records.
Station Cashier	Cashier Booth PC or Station Tablet	PIN / Password	Audit attendant remittances side-by-side against POS printouts; flag unresolved shortages; perform end-of-day physical note denomination count; log drawer cash expenses.
Branch Manager	Manager Smartphone or Tablet	Individual Password	Assign attendants to nozzles and shifts; supervise tanker deliveries and dip before/after discharge; record daily 06:00 PM underground tank dips; approve branch operational expenses.
Senior & Directors	Executive Laptop / Desktop View	Corporate Password / 2FA	Authorise retail fuel prices; authorize registered credit customer limits; confirm cash deposits against commercial bank credit alerts; decide attendant salary shortage deductions; view 5-station P&L.

4. Operational Daily Cycle & Mathematical Formulas

A. Expected Shift Sales Value Formula (\$2.10)

The system computes the monetary value of fuel dispensed based on mechanical meter dials:

```

Litres Sold = Closing Meter Reading - Opening Meter Reading
Nozzle Sales Value = Litres Sold × Active Unit Price
Total Expected Shift Sales = ∑ (Nozzle Sales Value across all assigned nozzles)

```

Mid-Shift Price Change Periodization (\$2.9): When Senior authorizes a price adjustment during an active shift, the system captures an instant meter snapshot (\$M_{switch}\$), splitting sales mathematically:

```

Expected Sales = ((M_switch - M_open) × Price_Old) + ((M_close - M_switch) × Price_New)

```

B. Attendant Shift Remittance & Variance Attribution (\$4.6, \$4.7)

Declared funds across four payment routes plus verified credit must equal the expected sales value:

```

Money Declared = Cash + POS Card Swipes + Station POS Transfers + Company Bank Transfers
Attendant Variance = (Money Declared + Verified Credit Sales) - Total Expected Shift Sales

```

Strict Accounting Rule: Software figures are never altered to artificially force a balance. Shortfalls (\$Variance < 0\$) are locked directly into the attendant's salary deduction register. Excesses (\$Variance > 0\$) are preserved as company payable to staff.

C. The Physical Cash Drawer Equation (§5.3)

Physical cash in the cashier's safe is audited at close of business using denomination counting:

Expected Closing Cash = Opening Cash + Cash Receipts - Physical Cash Expenses - Handed Over for Deposit

Counted Cash = $\sum$ (Denomination Value $\times$ Count of Notes for ₦1000, ₦500, ₦200, ₦100, ₦50, ₦20, ₦10)

Cash Variance = Counted Cash - Expected Closing Cash

Separation of Cash vs Bank (§5.2): Operating expenses paid via bank transfer are strictly excluded from cash drawer deductions. Cash handed over to couriers for commercial bank deposit remains held in an [Awaiting bank] state until Senior verifies the bank credit alert.

5. Anti-Fraud & Revenue Leakage Prevention Engine

The software enforces five strict systemic controls to eliminate the primary sources of fuel retail shrinkage:

Shrinkage Threat	Traditional Forecourt Vulnerability	Ajadico Systemic Countermeasure
Fake Bank Alerts & Transfer Slips	Customers show fake debit SMS alerts on their phone. Attendants accept them and leave station with a cash shortfall.	Uploading a payment screenshot never confirms funds. Two-tier verification: Cashier matches terminal printout; Senior matches corporate bank alert before funds are confirmed.
Clerical Shortage Messaging	Attendants or cashiers alter closing meter figures or claim higher opening dials to make accounts balance on paper.	Previous shift closing readings automatically lock as the next shift's opening baseline. Overrides require manager PIN with permanent historical audit trail (\$2.5).
Informal Forecourt Credit	Attendants dispense fuel to associates or transport operators on informal promises to pay later.	Credit sales can only be logged against accounts pre-authorized by Senior. Captures mandatory vehicle registration, driver name, litres, and signed requisition upload.
Tanker Discharge Shrinkage	Delivery drivers claim a full 33,000L discharge while withholding volume in compartments or suffering transit loss.	Mandatory Before & After physical tank dip measurements compute exact received litres. The variance against stated waybill litres is automatically logged.
False Shortages from Calibration	Pump meter calibration tests or generator diesel withdrawals are counted as missing sales.	Screen 07 logs 20L check measure can pour-backs and internal generator fuel, updating calculated tank stock without generating cash shortage.

6. Offline-First Forecourt Resilience Strategy

Forecourt operations in Nigeria cannot depend on 100% internet reliability. The mobile application incorporates an offline-first architecture:

- **Local Encrypted SQLite Storage:** Shift assignments, nozzle baselines, and active prices are cached locally on the tablet.
- **Offline PIN Authentication:** Attendant PIN hashes are verified locally, allowing instant login even when the cellular network is down.
- **Local Image Staging:** Photos of POS receipts and driver requisitions are compressed and staged on local flash storage.
- **Idempotent Background Synchronization:** When Wi-Fi/cellular connectivity restores, the background sync queue flushes records using immutable UUIDs, preventing duplicate submissions if a connection drops mid-upload (\$9.6).

7. Complete 17-Screen Specification Matrix

01. Shared Tablet PIN Login

Attendant / Cashier

Purpose: Fast station tablet authentication without keyboard typing (\$2.1–\$2.4).

UI Features: Grid of on-duty staff name cards; 60px numeric touch keypad (1–9, Clear, 0, Log in); PIN masking; Forgot PIN manager escalation link.

02. Attendant Shift Home

Pump Attendant

Purpose: Shift startup checklist and assigned nozzle oversight (\$2.2, \$2.6).

UI Features: Assigned nozzle cards (PMS/AGO) with supplying tank codes; carry-forward opening confirmation chip (Confirmed / Confirm opening); shift progress checklist; quick actions.

04. Closing Meter Readings

Pump Attendant

Purpose: Real-time fuel volume and sales value calculation at handover (\$2.5, \$2.10).

UI Features: Opening reading (readonly); closing reading decimal input; unit price (readonly); live litres sold indicator; live sales value (₹); total expected sales sum card.

05. Shift Remittance Declaration

Pump Attendant

Purpose: Multi-channel sales declaration and proof upload (\$4.1–\$4.7).

UI Features: Cash, POS Card, Station POS Transfer, Company Bank Transfer inputs; camera upload dropzone for POS merchant slips; credit deduction display; live color-coded difference.

06. Pump Credit Sale Entry

Pump Attendant

Purpose: Mid-shift fuel dispensing against approved credit accounts (\$4.8–\$4.10).

UI Features: Customer dropdown with running debt balance; nozzle picker; litres input; vehicle number; driver name; signed requisition photo capture; live total credit value.

07. Fuel Return & Calibration

Attendant / Manager

Purpose: Calibration pour-backs and internal generator fuel logging (\$3.4, \$3.5).

UI Features: Entry type selector (20L Calibration Pour-back, Generator Diesel, Company Vehicle); nozzle used; litres withdrawn; reason field; measure can photo attachment.

09. Cashier Verification Queue

Cashier / Manager

Purpose: Side-by-side reconciliation of meter sales vs physical payment slips (\$4.2–\$4.4).

UI Features: Left column: pump meter summary & declared payment checklist; Right column: POS receipt image carousel; Cashier comments; Actions: Verify, Flag Unresolved, Return to Staff.

10. Daily Physical Cash Count

Station Cashier

Purpose: Close-of-business physical note audit and cash drawer formula check (\$5.3–\$5.5).

UI Features: Denomination breakdown table (₦1000 down to ₦10); Expected cash equation display; Counted vs Expected variance; [Awaiting bank] holding indicator.

11. Record Branch Expense

Cashier / Manager

Purpose: Station operational expenditure logging (\$5.1, \$5.2).

UI Features: Category picker (generator maintenance, cleaning, utilities, stationery); amount input; payment source toggle (Sales Cash vs Bank Transfer); description; receipt photo.

12. Daily Tank Dip Audit

Branch Manager

Purpose: Physical underground inventory measurement vs book stock (\$3.2, \$3.3).

UI Features: Tank cards (T1 PMS, T2 PMS, T3 AGO); tank capacity; calculated book stock; physical dip entry; automatic discrepancy calculation with color-coded variance tags.

13. Fuel Delivery Reception

Branch Manager

Purpose: Tanker truck discharge and waybill shortage/excess logging (\$3.7–\$3.9).

UI Features: Receiving tank selector; supplier name; waybill number; stated litres; Before Delivery dip; After Delivery dip; actual received litres; transit discrepancy; waybill photo.

17. Branch Manager Dashboard

Branch Manager

Purpose: Real-time operational overview and exception queue (\$6.1, \$6.2).

UI Features: KPI metric tiles (PMS sold, AGO sold, sales value, credit given, expenses, deposits pending); Needs Attention queue; Shift statuses (Morning, Evening); Action bar.

18. Price Authorization & Snapshot

Senior (Central Admin)

Purpose: Authorized retail fuel price adjustment with meter transition snapshots (\$2.8, \$2.9).

UI Features: Product selector; new price input; directive reason; mandatory transition meter reading capture across all affected nozzles to close old price and open new price.

19. Attendant Salary Ledger

Senior (Central Admin)

Purpose: Shift discrepancy review and payroll deduction authorization (\$4.6, \$4.7).

UI Features: Unresolved shift shortage/excess register; attendant name; station; shift date; amount; Senior decision buttons: Approve Salary Deduction, Waive Discrepancy.

20. Bank Deposit Approvals

Senior (Central Admin)

Purpose: Matching cash handovers against commercial bank credit alerts (\$4.3, \$5.5).

UI Features: Station deposit queue; bank name; cashier name; amount; GTBank/Zenith/FirstBank alert matching button (moves funds from Awaiting Bank to Confirmed Bank Assets).

21. Credit Customer Ledger

Manager / Senior

Purpose: Central directory of authorized credit accounts and debt aging (\$4.8–\$4.10).

UI Features: Search filter; customer table (Outstanding balance, last repayment, due date, status tags); Senior-gated "Add Customer" authorization trigger.

22. Company Consolidated Reports

Senior & Directors

Purpose: 5-station monthly P&L and executive profitability dashboard (\$6.1, \$6.4).

UI Features: Multi-branch summary cards (Total Revenue, Gross Profit, Operating Expenses, Net Profit); 5-station comparative breakdown table with PMS/AGO litres, COGS, and margins.

8. Supabase & PostgreSQL Production Database Architecture

The backend database schema in `supabase/schema.sql` provides strict transactional integrity, foreign key cascades, and row-level security:

Table Name	Primary Keys & Relationships	Key Data Columns & Generated Fields
stations	id (UUID PK)	code, name, address, state, manager_name, phone, has_interlocked_tanks
fuel_products	id (UUID PK)	code ('PMS', 'AGO'), name, unit ('Litres')
fuel_prices	id (UUID PK) FK station_id, product_id	price_per_litre, effective_from, effective_to, authorised_by
tanks	id (UUID PK) FK station_id, product_id	code, capacity_litres, current_dip_litres, calculated_stock_litres, is_interlocked
pumps & nozzles	id (UUID PK) FK pump_id, tank_id	nozzle_number, latest_meter_reading, is_active
profiles	id (UUID PK) FK station_id	display_name, full_name, role (enum), pin_hash, phone, is_active
shifts	id (UUID PK) FK station_id	business_date, shift_type ('morning', 'afternoon_evening'), status
shift_nozzle_assignments	id (UUID PK) FK shift_id, nozzle_id	opening_reading, closing_reading, price_per_litre, litres_sold (GENERATED), expected_sales_value (GENERATED)
remittances	id (UUID PK) FK shift_id, attendant_id	cash_declared, pos_card_declared, pos_transfer_declared, bank_transfer_declared, credit_sales_declared, total_declared (GENERATED), variance (GENERATED)
daily_cash_counts	id (UUID PK) FK station_id, cashier_id	opening_cash, cash_receipts, cash_expenses, handed_over_for_deposit, expected_closing_cash, counts for ₦1000..₦10, variance (GENERATED)
credit_customers & sales	id (UUID PK) FK station_id	company_name, credit_limit, outstanding_balance, payment_terms_days, status, vehicle_number, driver_name
attendant_salary_adjustments	id (UUID PK) FK attendant_id, shift_id	amount, adjustment_type ('shortage', 'excess'), status, decided_by, decision_notes

9. BRD Section 7: Key Open Business Decisions

Section 7 of BRD v3 identified four operational questions requiring confirmation before production deployment:

1. Interlocked Tanks Protocol (Ikeja & Surulere):

Implemented Solution: When switching from Tank 1 to Tank 2 mid-shift, the app prompts for the switch time, meter readings on the connected nozzles, and physical dips on both tanks. Litres dispensed before the switch timestamp are deducted from Tank 1; subsequent litres are deducted from Tank 2.

2. Delivery Measurement During Active Dispensing:

Implemented Solution: When receiving fuel, if dispensing continues on that tank, the delivery form logs the nozzle opening meters at tanker connection and closing meters at tanker disconnect: $\text{Actual Received} = (\text{Dip After} - \text{Dip Before}) + \text{Litres Dispensed During Discharge}$.

3. Informal Credit & Salary Deduction Thresholds:

Implemented Solution: Shift shortages under ₦5,000 are flagged for routine payroll deduction; shortages exceeding ₦5,000 trigger an immediate alert to Senior for formal authorization or waiver.

4. Month-End Executive Profit Layout:

Implemented Solution: Fuel purchases are treated as current inventory assets. COGS is recognized on nozzle sales using weighted average cost, producing accurate Gross Margin and Net Operating Profit after branch expenses.